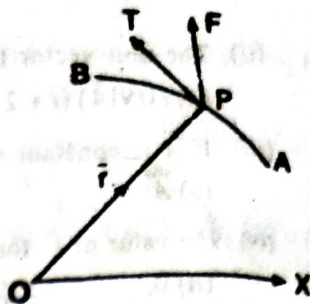


5.40 INTEGRATION OF VECTORS

LINE INTEGRAL

Let $\vec{F}(x, y, z)$ be a vector function and a curve AB .

Line integral of a vector function $\vec{F}$ along the curve AB is defined as integral of the component of $\vec{F}$ along the tangent to the curve AB .



Component of $\vec{F}$ along a tangent PT at P

= Dot product of $\vec{F}$ and unit vector along PT

$$= \vec{F} \cdot \frac{d\vec{r}}{ds} \quad \left(\frac{d\vec{r}}{ds} \text{ is a unit vector along tangent } PT \right)$$

Line integral = $\int_A^B \vec{F} \cdot \frac{d\vec{r}}{ds}$ from A to B along the curve

$$\therefore \text{Line integral} = \int_c \left(\vec{F} \cdot \frac{d\vec{r}}{ds} \right) ds = \int_c \vec{F} \cdot d\vec{r}$$

Note (1) Work. If $\vec{F}$ represents the variable force acting on a particle along arc AB , then the total work done = $\int_A^B \vec{F} \cdot d\vec{r}$

(2) Circulation. If $\vec{F}$ represents the velocity of a liquid then $\oint_c \vec{F} \cdot d\vec{r}$ is called the circulation of F round the curve c .

If $\oint \vec{F} \cdot d\vec{r} = 0$, then, the field F is called conservative, i.e. no work is done and the energy is conserved.

(3) When the path of integration is a closed curve then notation of integration is $\oint$ in place of $\int$.

Example 89. If a force $\vec{F} = 2x^2y\mathbf{i} + 3xy\mathbf{j}$ displaces a particle in the xy -plane from $(0, 0)$ to $(1, 4)$ along a curve $y = 4x^2$. Find the work done.

Solution. Work done = $\int_c \vec{F} \cdot d\vec{r}$ $\vec{r} = x\mathbf{i} + y\mathbf{j}$
 $d\vec{r} = dx\mathbf{i} + dy\mathbf{j}$

$$= \int_c (2x^2y\mathbf{i} + 3xy\mathbf{j}) \cdot (dx\mathbf{i} + dy\mathbf{j})$$

$$= \int_c (2x^2y dx + 3xy dy)$$

Putting the value of y and dy we get $\begin{cases} y = 4x^2 \\ dy = 8x dx \end{cases}$

$$= \int_0^1 [2x^2(4x^2) dx + 3x(4x^2) 8x dx]$$

$$= 104 \int_0^1 x^4 dx = 104 \left(\frac{x^5}{5} \right)_0^1 = \frac{104}{5} \quad \text{Ans.}$$

Example 90. Compute $\int_c \vec{F} \cdot d\vec{r}$, where $\vec{F} = \frac{iy - jx}{x^2 + y^2}$ and c is the circle $x^2 + y^2 = 1$ traversed counter clockwise.

Solution. $\vec{r} = ix + jy + kz$, $d\vec{r} = i dx + j dy + k dz$

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C \frac{iy - jx}{x^2 + y^2} \cdot (idx + jdy + kdz) \\ &= \int_C \frac{ydx - xdy}{x^2 + y^2} = \int_C (ydx - xdy) \quad [\because x^2 + y^2 = 1]\end{aligned}$$

Parametric equation of the circle are $x = 1 \cos \theta$, $y = 1 \sin \theta$.

Putting $x = \cos \theta$, $y = \sin \theta$, $dx = -\sin \theta d\theta$, $dy = \cos \theta d\theta$

$$\begin{aligned}&= \int_0^{2\pi} \sin \theta (-\sin \theta d\theta) - \cos \theta (\cos \theta d\theta) \\ &= -\int_0^{2\pi} (\sin^2 \theta + \cos^2 \theta) d\theta = -\int_0^{2\pi} d\theta \\ &= -\left(\theta\right)_0^{2\pi} = -2\pi\end{aligned}$$

Ans.

Example. 91. If $\mathbf{F} = \nabla \phi$ show that the work done in moving a particle in the force field $\mathbf{F}$ from $A(x_1, y_1, z_1)$ to $B(x_2, y_2, z_2)$ is independent of the path joining the two points.

Solution

$$\begin{aligned}\text{Work done} &= \int_A^B \mathbf{F} \cdot d\mathbf{r} = \int_A^B \nabla \phi \cdot d\mathbf{r} \\ &= \int_A^B \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \phi \cdot (idx + jdy + kdz) \\ &= \int_A^B \left(i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z} \right) \cdot (idx + jdy + kdz) \\ &= \int_A^B \left(\frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz \right) \\ &= \int_A^B d\phi = \left[\phi \right]_A^B = \phi(B) - \phi(A) \\ &= \phi(x_2, y_2, z_2) - \phi(x_1, y_1, z_1)\end{aligned}$$

Hence the work done depends only on the end points A and B not on the path from A to B .

Note : If $\int_A^B \mathbf{F} \cdot d\mathbf{r}$ is to be proved to be independent of path, then $\mathbf{F} = \nabla \phi$

Here $\mathbf{F}$ is called conservative (irrotational) vector field and ϕ is called the scalar potential.

And $\nabla \times \mathbf{F} = \nabla \times \nabla \phi = 0$

Example 92. Show that the integral

$\int_{(1,2)}^{(3,4)} (xy^2 + y^3) dx + (x^2y + 3xy^2) dy$ is independent of the path joining the points $(1, 2)$ and $(3, 4)$. Hence, evaluate the integral.

Solution. $\int_{(1,2)}^{(3,4)} (xy^2 + y^3) dx + (x^2y + 3xy^2) dy$

$$= \int_{(1,2)}^{(3,4)} [(xy^2 + y^3) i + (x^2y + 3xy^2) j] \cdot (idx + jdy)$$

$$= \int_{(1,2)}^{(3,4)} \vec{F} \cdot d\vec{r}$$

$$\nabla \times \vec{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2 + y^3 & x^2y + 3xy^2 & 0 \end{vmatrix}$$

$$= (0 - 0)i + (0 - 0)j + (2xy + 3y^2 - 2xy - 3y^2)k$$

$$= 0$$

If $\nabla \times \vec{F} = 0$ then $\vec{F} = \nabla \phi$

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy$$

$$= \left(\frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j \right) \cdot (i dx + j dy) = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} \right) \phi \cdot (i dx + j dy)$$

$$= \nabla \phi \cdot d\vec{r} = \vec{F} \cdot d\vec{r}$$

$$= (xy^2 + y^3) dx + (x^2y + 3xy^2) dy$$

$$= [(x dx) y^2 + x^2 (y dy)] + [y^3 dx + x (3y^2 dy)]$$

$$= \frac{1}{2} d(x^2 y^2) + d(xy^3)$$

$$\phi = \int \left[\frac{1}{2} d(x^2 y^2) + d(xy^3) \right]$$

$$= \frac{1}{2} x^2 y^2 + xy^3$$

$$[\phi]_{(1,2)}^{(3,4)} = \left[\frac{1}{2} x^2 y^2 + xy^3 \right]_{(1,2)}^{(3,4)} = \left[\frac{1}{2} (9)(16) + (3)(64) \right] - \left[\frac{1}{2} (1)(4) + (1)(8) \right]$$

$$= 72 + 192 - 2 - 8 = 254$$

Example 93. Determine whether the line integral

Ans.

$\int_C (2xy z^2 dx + (x^2 z^2 + z \cos yz) dy + (2x^2 yz + y \cos yz) dz)$ is independent of the path of integration? If so, then evaluate it from $(1, 0, 1)$ to $(0, \frac{\pi}{2}, 1)$.

Solution. $\int_C (2xy z^2 dx + (x^2 z^2 + z \cos yz) dy + (2x^2 yz + y \cos yz) dz)$

$$= \int_C (2xy z^2 i + (x^2 z^2 + z \cos yz) j + (2x^2 yz + y \cos yz) k) \cdot (i dx + j dy + k dz)$$

$$= \int_C \vec{F} \cdot d\vec{r}$$

This integral is independent of path of integration if

$$\vec{F} = \nabla \phi \Rightarrow \nabla \times \vec{F} = 0$$

$$\nabla \times \vec{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy z^2 & x^2 z^2 + z \cos yz & 2x^2 yz + y \cos yz \end{vmatrix}$$

$= (2x^2z + \cos yz - yz \sin yz - 2x^2z - \cos yz + yz \sin yz) i - (4xyz - 4xyz) j + (2xz^2 - 2xz^2) k = 0$
Hence the line integral is independent of path.

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz \quad (\text{Total differentiation})$$

$$\begin{aligned} &= \left(i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z} \right) \cdot (i dx + j dy + k dz) = \nabla \phi \cdot d\mathbf{r} = \mathbf{F} \cdot d\mathbf{r} \\ &= [(2xyz^2 i + (x^2z^2 + z \cos yz) j + (2x^2yz + y \cos yz) k] \cdot (i dx + j dy + k dz) \\ &= 2xyz^2 dx + (x^2z^2 + z \cos yz) dy + (2x^2yz + y \cos yz) dz \\ &= [(2x dx) yz^2 + x^2 (dy) z^2 + x^2 y (2z dz)] + [(\cos yz dy) z + (\cos yz dz) y] \\ &= d(x^2yz^2) + d(\sin yz) \end{aligned}$$

$$\phi = \int d(x^2yz^2) + \int d(\sin yz) = x^2yz^2 + \sin yz$$

$$\left[\phi \right]_A^B = \phi(B) - \phi(A) = [x^2yz^2 + \sin yz]_{(0, \frac{\pi}{2}, 1)} - [x^2yz^2 + \sin yz]_{(1, 0, 1)}$$

$$= \left[0 + \sin \left(\frac{\pi}{2} \times 1 \right) \right] - [0 + 0] = 1$$

Ans.

Exercise 5.19

- Find the work done by a force $y\mathbf{i} + x\mathbf{j}$ which displace a particle from origin to a point $(i + j)$. Ans. 1
- A vector field is given by $\vec{F} = (x^2 - y^2 + x)\hat{i} - (2xy + y)\hat{j}$. show that the field is irrotational and find its scalar potential. Hence evaluate line integral $\int \vec{F} \cdot d\vec{r}$ from $(1, 2)$ to $(2, 1)$. Ans. $-\frac{35}{6}$
(A.M.I.E.T.E., Summer 2003)
- Show that $\mathbf{V} = (2xy + z^2)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$, is a conservative field. Find its scalar potential ϕ such that $\mathbf{V} = \text{grad } \phi$. Find the work done by the force $\mathbf{V}$ in moving a particle from $(1, -2, 1)$ to $(3, 1, 4)$.
Ans. $x^2y + xz^3$, 202

- Show that the line integral $\int_c (2xy + 3) dx + (x^2 - 4z) dy - 4y dz$ where c is any path joining $(0, 0, 0)$ to $(1, -1, 3)$ does not depend on the path c and evaluate the line integral.

Ans. 14

- Find the work done in moving a particle once round the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1, z = 0$ under the field of force given by $\mathbf{F} = (2x - y + z)\mathbf{i} + (x + y - z^2)\mathbf{j} + (3x - 2y + 4z)\mathbf{k}$, is the field of force conservative?

(A.M.I.E.T.E., Winter 2000) Ans. 40π

- Find the work done by a force $\vec{F} = \sin y \hat{i} + x(1 + \cos y)\hat{j} + z\hat{k}$ by moving a particle once around the circle $x^2 + y^2 = a^2$.
(A.M.I.E.T.E., Dec. 2004)

- Show that the vector field $\vec{F} = (yze^{xz} - 4x)\hat{i} + (xze^{xz} + z)\hat{j} + (xye^{xz} + y)\hat{k}$ is conservative. Hence evaluate the line integral $\int (yze^{xz} - 4x) dx + (xze^{xz} + z) dy + (xye^{xz} + y) dz$ along a path joining the points $(0, 0, 0)$ to $(1, 1, 1)$.
(A.M.I.E.T.E., Dec. 2004)

- A force field $\vec{F}$ is said to be conservative if
(a) $\text{Curl } \vec{F} = 0$ (b) $\text{grad } \vec{F} = 0$ (c) $\text{Div } \vec{F} = 0$ (d) $\text{Curl } (\text{grad } \vec{F}) = 0$ (A.M.I.E.T.E., Dec. 2006) Ans. (d)

5.41. SURFACE INTEGRAL

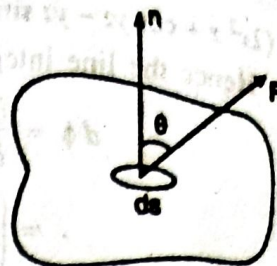
Let $\vec{F}$ be a vector function and S be the given surface.

Surface integral of a vector function $\vec{F}$ over the surface S is defined as the integral of the components of $\vec{F}$ along the normal to the surface.

Component of $\vec{F}$ along the normal
 $= \vec{F} \cdot \vec{n}$ where $\vec{n}$ is the unit normal vector to an element ds and

$$\hat{n} = \frac{\text{grad } f}{|\text{grad } f|}$$

$$ds = \frac{dx dy}{(\hat{n} \cdot \vec{k})}$$



$$\text{Surface integral of } \vec{F} \text{ over } S = \Sigma \vec{F} \cdot \hat{n} = \iint_S (\vec{F} \cdot \hat{n}) ds$$

Note. (1) Flux $= \iint_S (\vec{F} \cdot \hat{n}) ds$ where $\vec{F}$ represents the velocity of a liquid.

If $\iint_S (\vec{F} \cdot \hat{n}) ds = 0$, then $\vec{F}$ is said to be a solenoidal vector point function.

Example 94. Show that $\iint_S \vec{F} \cdot \hat{n} ds = \frac{3}{2}$, where

$\vec{F} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$ and S is the surface of the cube bounded by the planes,

$$x = 0, x = 1, y = 0, y = 1, z = 0, z = 1.$$

Solution.

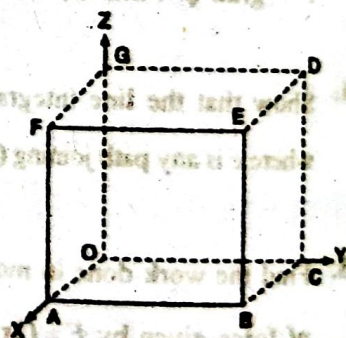
S. No.	Surface	Outward Normal	ds	Eq. of surface
1	OABC	$-\vec{k}$	$dx dy$	$z = 0$
2	DEFG	$\vec{k}$	$dx dy$	$z = 1$
3	OAFG	$-\vec{j}$	$dx dz$	$y = 0$
4	BCDE	$\vec{j}$	$dx dz$	$y = 1$
5	ABEF	$\vec{i}$	$dy dz$	$x = 1$
6	OCDG	$-\vec{i}$	$dy dz$	$x = 0$

$$\iint_S \vec{F} \cdot \hat{n} ds = \iint_{OABC} \vec{F} \cdot \hat{n} ds$$

$$+ \iint_{DEFG} \vec{F} \cdot \hat{n} ds + \iint_{OAFG} \vec{F} \cdot \hat{n} ds$$

$$+ \iint_{BCDE} \vec{F} \cdot \hat{n} ds + \iint_{ABEF} \vec{F} \cdot \hat{n} ds$$

$$+ \iint_{OCDG} \vec{F} \cdot \hat{n} ds \quad \dots (1)$$



$$\iint_{OABC} \vec{F} \cdot \hat{n} ds = \iint_{OABC} (4xz\vec{i} - y^2\vec{j} + yz\vec{k}) \cdot (-\vec{k}) dx dy = \int_0^1 \int_0^1 -yz dx dy = 0 \quad (\text{as } z = 0)$$

$$\iint_{DEFG} (4xz\vec{i} - y^2\vec{j} + yz\vec{k}) \cdot \vec{k} dx dy = \iint_{DEFG} yz dx dy = \int_0^1 \int_0^1 y(1) dx dy$$

$$= \int_0^1 dx \left[\frac{y^2}{2} \right]_0^1 = \left[x \right]_0^1 \frac{1}{2} = \frac{1}{2}$$

$$\iint_{OAFG} (4xz\vec{i} - y^2\vec{j} + yz\vec{k}) \cdot (-\vec{j}) dx dz = \iint_{OAFG} -y^2 dx dz = 0 \quad (\text{as } y = 0)$$

$$\iint_{BCDE} (4xz\vec{i} - y^2\vec{j} + yz\vec{k}) \cdot \vec{j} dx dz = \iint_{BCDE} (-y^2) dx dz$$

$$= - \int_0^1 dx \int_0^1 dz = - (x)_0^1 (z)_0^1 = -1 \quad (\text{as } y=1)$$

$$\begin{aligned} \iint_{ABEF} (4xz i - y^2 j + yz k) \cdot i \, dy \, dz &= \iint 4xz \, dy \, dz = \int_0^1 \int_0^1 4(1)z \, dy \, dz \\ &= 4(y)_0^1 \left(\frac{z^2}{2} \right)_0^1 = 4(1) \left(\frac{1}{2} \right) = 2 \end{aligned}$$

$$\iint_{OCDE} (4xz i - y^2 j + yz k) \cdot (-i) \, dy \, dz = \int_0^1 \int_0^1 -4xz \, dy \, dz = 0 \quad (\text{as } x=0)$$

On putting these values in (1), we get

$$\iint_S \vec{F} \cdot \hat{n} \, ds = 0 + \frac{1}{2} + 0 - 1 + 2 + 0 = \frac{3}{2} \quad \text{Proved}$$

5.42. VOLUME INTEGRAL

Let $\vec{F}$ be a vector point function and volume V enclosed by a closed surface.

The volume integral = $\iiint_V \vec{F} \, dv$

Example 95. If $\vec{F} = 2z i - xj + yk$, evaluate $\iiint_V \vec{F} \, dv$ where v is the region bounded by the surfaces

$$x = 0, y = 0, x = 2, y = 4, z = x^2, z = 2.$$

Solution. $\iiint_V \vec{F} \, dv = \iiint (2z i - xj + yk) \, dx \, dy \, dz$

$$= \int_0^2 dx \int_0^4 dy \int_{x^2}^2 (2z i - xj + yk) \, dz$$

$$= \int_0^2 dx \int_0^4 dy [z^2 i - xzj + yzk]_{x^2}^2$$

$$= \int_0^2 dx \int_0^4 dy [4i - 2xzj + 2yzk - x^4i + x^3j - x^2yk]$$

$$= \int_0^2 dx \left[4yi - 2xyj + y^2k - x^4yi + x^3yj - \frac{x^2y^2}{2}k \right]_0^4$$

$$= \int_0^2 (16i - 8xj + 16k - 4x^4i + 4x^3j - 8x^2k) \, dx$$

$$= \left[16xi - 4x^2j + 16xk - \frac{4x^5}{5}i + x^4j - \frac{8x^3}{3}k \right]_0^2$$

$$= 32i - 16j + 32k - \frac{128}{5}i + 16j - \frac{64}{3}k$$

$$= \frac{32i}{5} + \frac{32k}{3} = \frac{32}{15}(3i + 5k) \quad \text{Ans}$$

5.43. GREEN'S THEOREM (For a plane)

Statement. If $\phi(x, y)$, $\psi(x, y)$, $\frac{\partial \phi}{\partial y}$ and $\frac{\partial \psi}{\partial x}$ be continuous functions over a region

bounded by simple closed curve C in $x-y$ plane, then

$$\oint_C (\phi \, dx + \psi \, dy) = \iint_R \left(\frac{\partial \psi}{\partial x} - \frac{\partial \phi}{\partial y} \right) \, dx \, dy$$